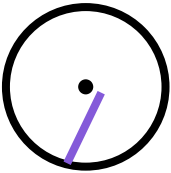
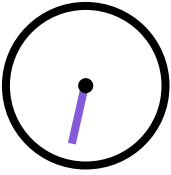
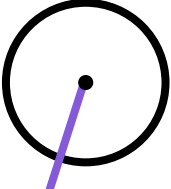




Multiplicative relationships in circles

Task A: Parts of a circle

1) For each diagram, explain why the length of the line segment is not the **radius**.

- a)  *The line segment does not go to the centre of the circle.*
- b)  *The line segment does not go to the edge of the circle.*
- c)  *The line segment goes beyond the edge of the circle.*

2) Fill in the blanks with numbers.

- a) The **diameter** of a circle is 2 times the **radius**.
- b) A circle with a **radius** of 6 cm has a **diameter** of 12 cm.
- c) A circle with a **diameter** of 6 cm has a **radius** of 3 cm.
- d) The distance from the centre of a circle to its edge is 10 m.
Its **radius** is 10 m and its **diameter** is 20 m.
- e) The furthest distance between two points on the edge of a circle is 24 m.
Its **radius** is 12 m and its **diameter** is 24 m.
-

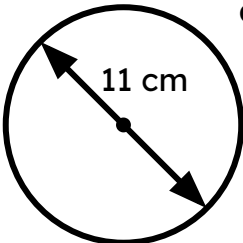
Task B: The relationship between circumference and diameter

1) Fill in the blanks in the table.

radius	0.5	1	3	5	12	50
diameter	1	2	6	10	24	100
circumference	π	2π	6π	10π	24π	100π

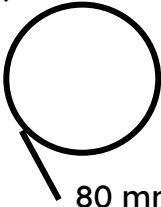
2) Find the missing length in each circle, giving your answer to 3 significant figures.

a) **diameter : circumference**



11 cm : 34.6 cm

b) **diameter : circumference**



25.5 mm : 80 mm

